



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

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- Methodology
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- Conclusion
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Executive Summary

- Summary of methodologies
 - Data Collection through API
 - Data Collection with Web Scraping
 - Data Wrangling
 - Exploratory Data Analysis with SQL
 - Exploratory Data Analysis with Data Visualization
 - Interactive Visual Analytics with Folium
 - Machine Learning Prediction
- Summary of all results
 - Exploratory Data Analysis result
 - Interactive analytics in screenshots
 - Predictive Analytics result

Executive Summary

- **Commercial Objective:** Determine whether SpaceX will successfully land its Falcon 9 first-stage booster for any given mission to predict launch pricing advantages (\$62M reused vs. \$165M+ expendable).
- **Core Finding:** Landing success is primarily driven by orbital flight profiles and mission experience rather than just payload weight. High-energy missions (GTO) still pose the highest recovery risks.
- **Model Confidence:** Our predictive models forecast recovery outcomes with 83.33% accuracy on unseen flight data.
- **Strategic Takeaway:** Competitors must factor in SpaceX's near-guaranteed reusability for standard low-orbit missions, while focusing commercial bids on high-energy profiles where recovery margins remain narrow.

Introduction & Business Statement

- **The Cost Factor:** Reusable boosters allow SpaceX to price flights at roughly \$62 million, well below traditional single-use rockets costing over \$165 million.
 - **Business Impact:** If a competitor knows a booster cannot be recovered, they know SpaceX faces higher costs and can price launch bids more aggressively.
 - **Project Goal:** Build a classification model to predict whether a booster lands safely (Class = 1) or is lost (Class = 0).
-
- Problems you want to find answers
 - What key factors determine a successful rocket landing?
 - How do various features interact to influence the landing success rate?
 - What operating conditions are required to ensure a successful landing program?

Section 1

Methodology

Methodology

Data Collection (API & Scraping)

Data collection methodology:

- Data was collected using SpaceX API and web scraping from Wikipedia.
- Perform data wrangling
 - One-hot encoding was applied to categorical features
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

The data was collected using various methods

- **SpaceX REST API:** Extracted 90 historical launches containing payload mass, target orbit, core serials, and launch complexes.
- **Web Scraping:** Scraped Falcon 9 launch records from Wikipedia using BeautifulSoup to cross-validate early flight telemetry.
- **Output:** Standardized structured dataset capturing missions from early testing through operational cadence.

Data Collection – SpaceX API and Scraping

- The link to the notebook is https://github.com/onurtorun/IBM-Applied-Data-Science-Capstone/blob/main/1-1_Complete_the_Data_Collection_API_Lab/jupyter-labs-spacex-data-collection-api.ipynb

```
# Load pre-extracted SpaceX dataset directly from IBM Cloud Storage
dataset_url = 'https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_1.csv'
data_falcon9 = pd.read_csv(dataset_url)

# Display shape to confirm 90 rows and 17 columns
print("Data shape:", data_falcon9.shape)
data_falcon9.head()
```

Data shape: (90, 17)

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857

Data Cleaning & Wrangling

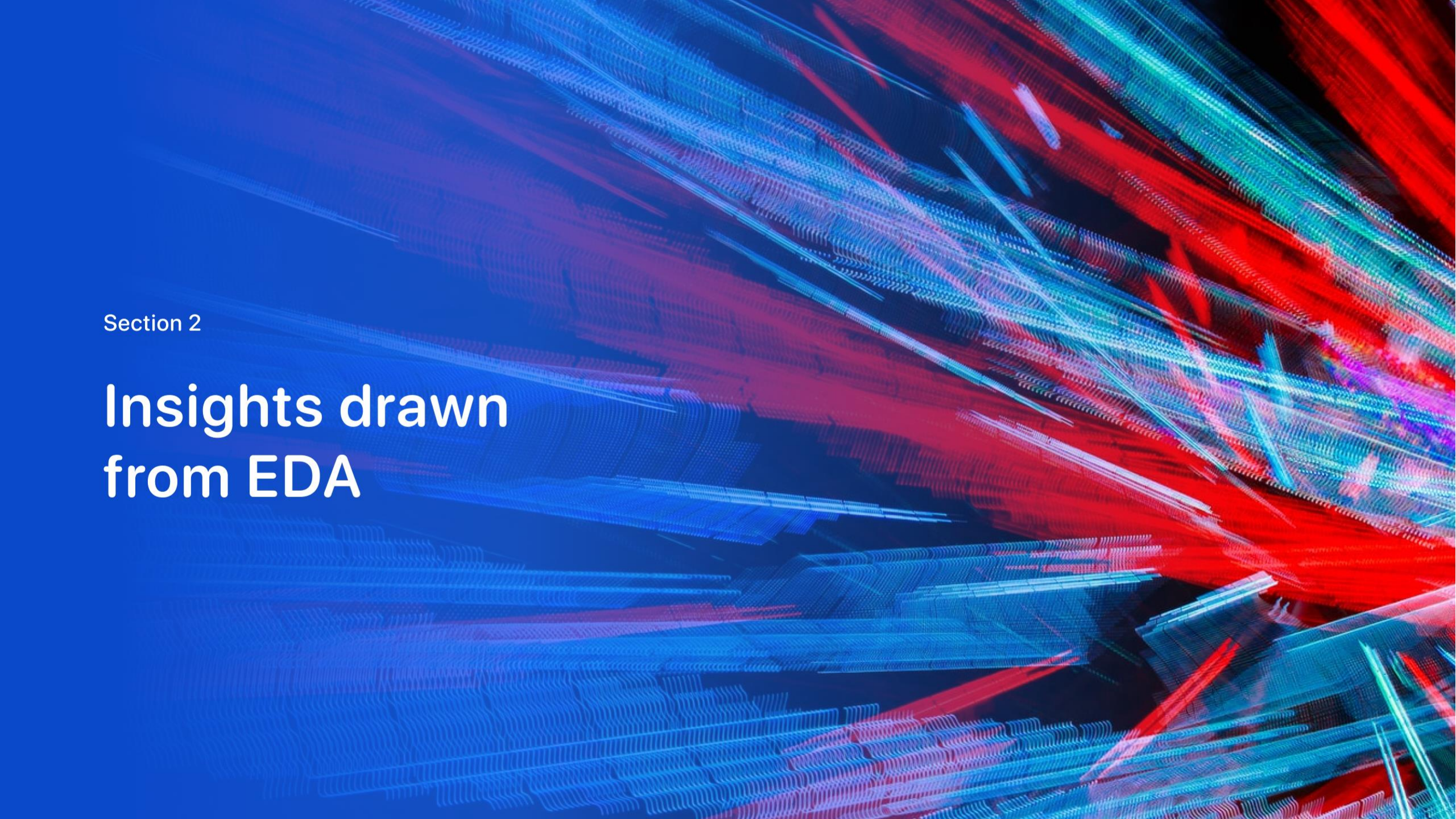
Missing Data: Addressed missing values in payload mass using class-based averages and handled missing landing pad assignments.

Outcome Labels: Mapped successful landings (ground pad, drone ship, calm ocean) to 1, and failed or expendable launches to 0.

Feature Set: Cleaned and structured key drivers including flight number, site location, orbit destination, and booster block version.

```
data_falcon9.isnull().sum()
```

FlightNumber	0
Date	0
BoosterVersion	0
PayloadMass	5
Orbit	0
LaunchSite	0
Outcome	0
Flights	0
GridFins	0
Reused	0
Legs	0
LandingPad	26
Block	0
ReusedCount	0
Serial	0
Longitude	0
Latitude	0
dtype:	int64

The background of the slide is a complex, abstract composition. It features a dark blue base color on the left side, which transitions into a vibrant, multi-colored area on the right. This transition area is filled with numerous thin, diagonal streaks in shades of red, orange, and yellow, creating a sense of motion and energy. Overlaid on these streaks is a faint, grid-like pattern of small, light-colored squares, reminiscent of a digital or data visualization theme.

Section 2

Insights drawn from EDA

Exploratory Data Analysis (SQL)

Setup: Queried the dataset using SQLite (my_data1.db) to identify core operational patterns.

Key Numbers: Found that CCAFS LC-40 hosted the majority of initial missions.

Milestones: Identified the first successful landing milestone and tracked early launch outcomes.

Booster Analysis: Verified maximum payload records across booster series and compared drone ship versus ground pad successes.

The link to the notebook is

https://github.com/onurtorun/IBM-Applied-Data-Science-Capstone/blob/main/2-1_Exploratory_Data_Analysis_with_SQL/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Task 7

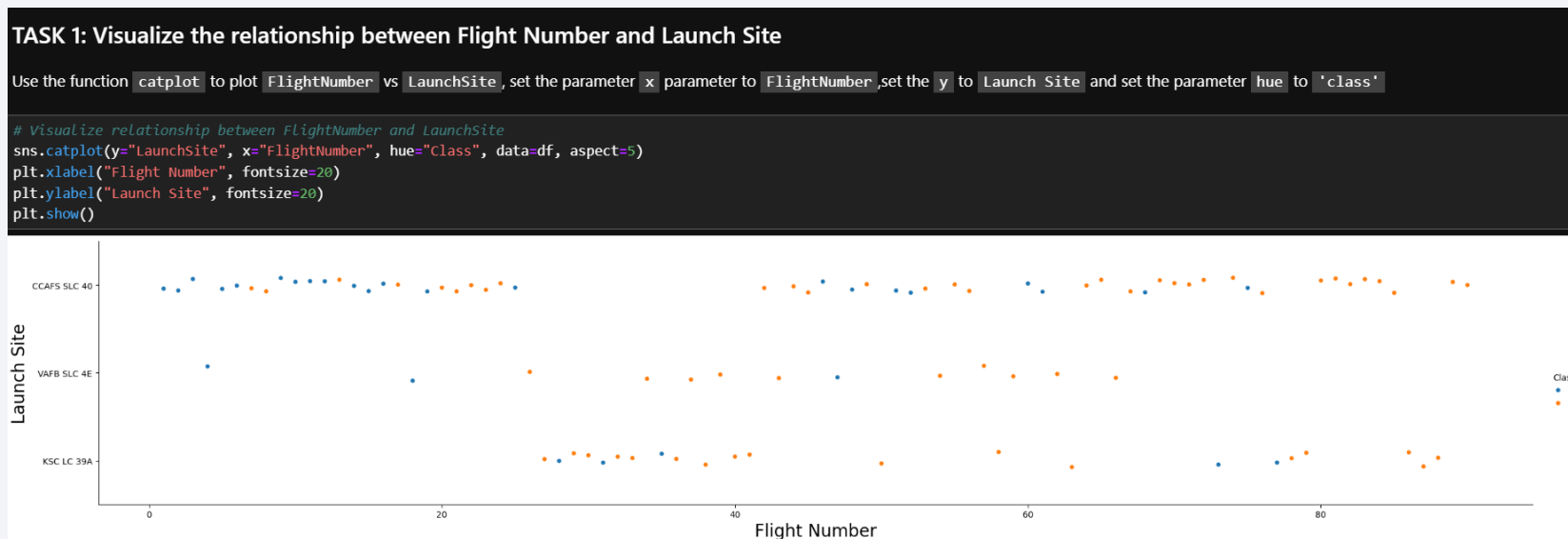
List the total number of successful and failure mission outcomes

```
%sql SELECT "Mission_Outcome", COUNT(*) AS Total_Count FROM SPACEXTABLE GROUP BY "Mission_Outcome";
* sqlite:///my_data1.db
Done.
```

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

EDA - Flight Number vs. Launch Site

- **Flight Growth:** CCAFS LC-40 carried almost all early test and commercial missions.
- **Expanding Operations:** KSC LC-39A and VAFB SLC-4E took on regular flight volume as the launch calendar grew.
- **Clear Trend:** Landing success improved markedly in later flight numbers across all launch complexes.



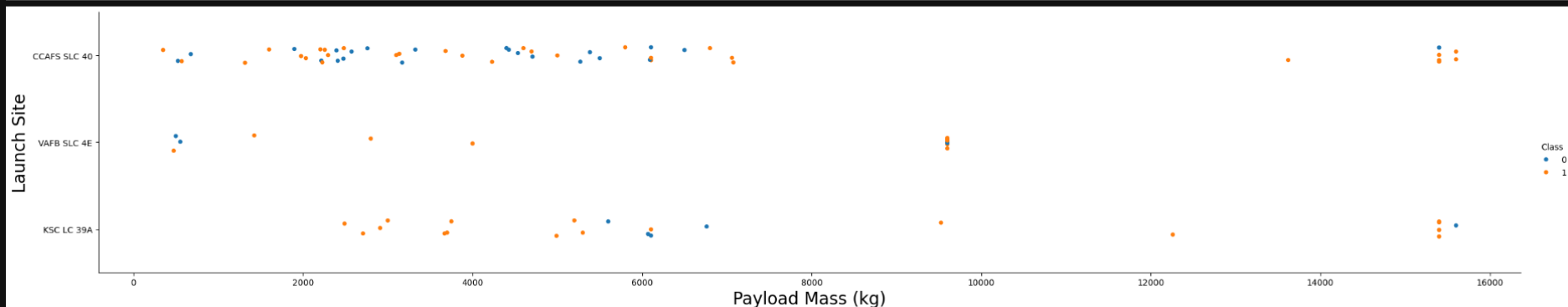
EDA - Payload Mass vs. Launch Site

- **Heavy Lift:** KSC LC-39A and CCAFS LC-40 handled the heaviest payloads, up to roughly 15,000 kg.
- **Polar Routes:** VAFB SLC-4E handled smaller payloads aimed at polar orbits.
- **Recovery Versatility:** Successful landings occurred across both light commercial satellites and maximum-weight missions.

TASK 2: Visualize the relationship between Payload Mass and Launch Site

We also want to observe if there is any relationship between launch sites and their payload mass.

```
# Visualize relationship between PayloadMass and LaunchSite
sns.catplot(y="LaunchSite", x="PayloadMass", hue="Class", data=df, aspect=5)
plt.xlabel("Payload Mass (kg)", fontsize=20)
plt.ylabel("Launch Site", fontsize=20)
plt.show()
```

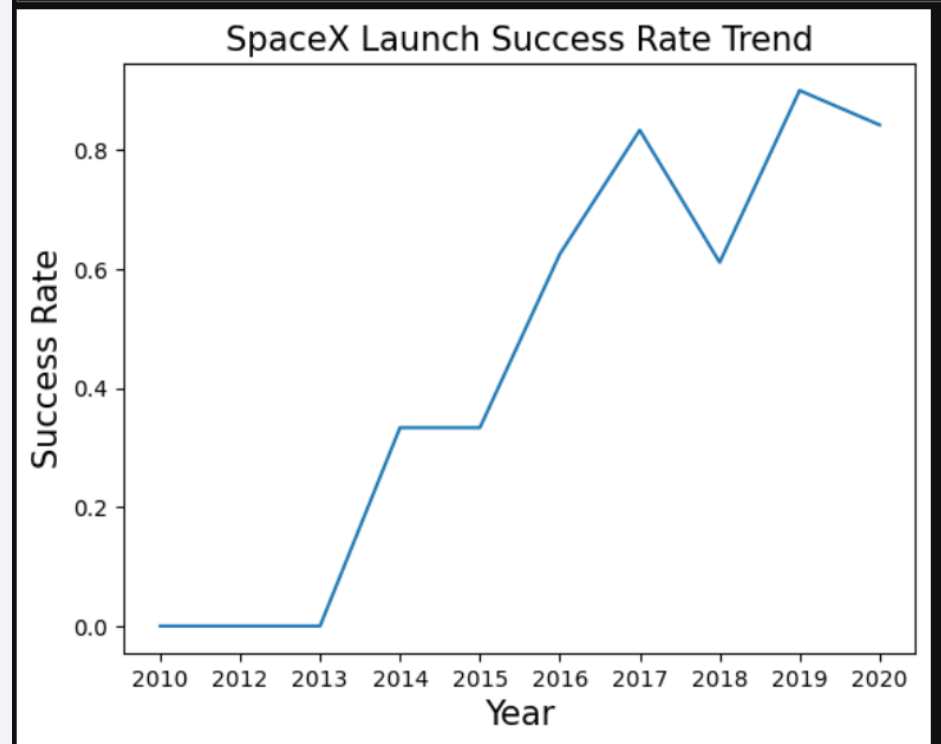


EDA - Yearly Success Rate Trend

- **2010 to 2013:** 0% recovery during early experimental flights.
- **Post-2015 Shift:** Landing rates improved quickly once grid fins, landing legs, and drone ship recovery stabilized.
- **2020 Standard:** Reached over 80% landing success, making reusability the normal operational standard.

```
# Calculate yearly average success rate
yearly_success = df.groupby('Date')['Class'].mean().reset_index()

# Plot launch success yearly trend
sns.lineplot(x="Date", y="Class", data=yearly_success)
plt.xlabel("Year", fontsize=15)
plt.ylabel("Success Rate", fontsize=15)
plt.title("SpaceX Launch Success Rate Trend", fontsize=15)
plt.show()
```



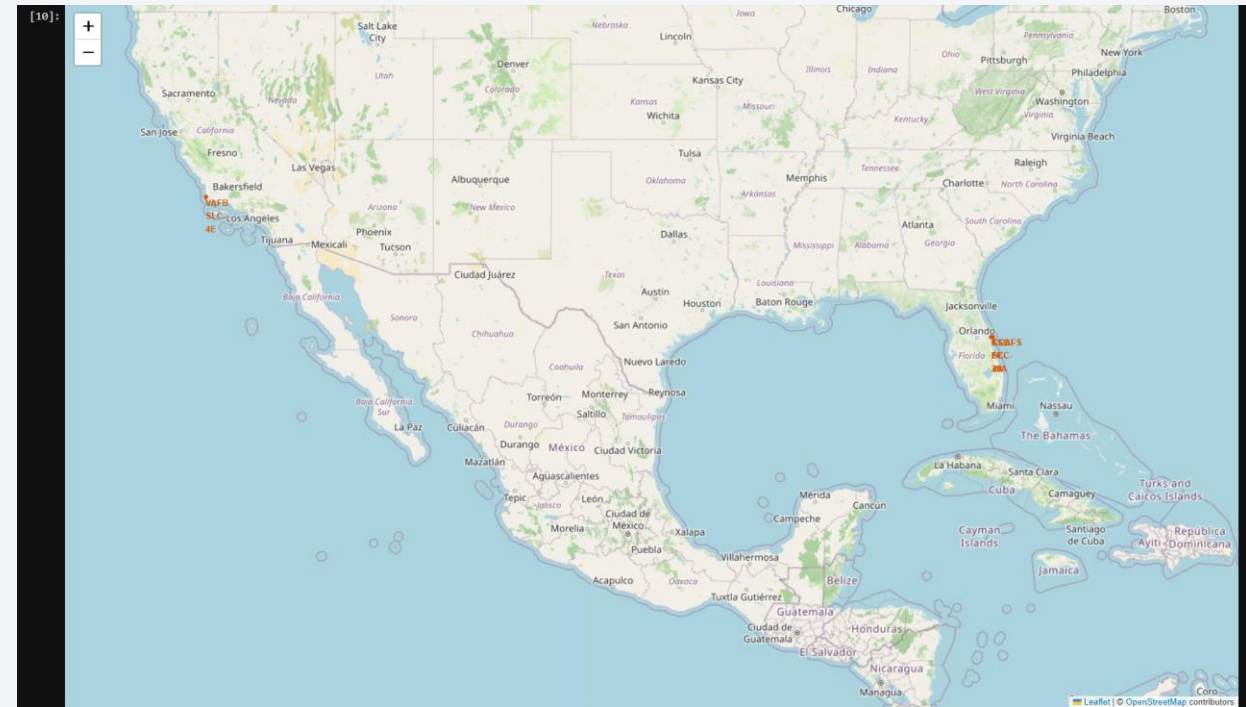
Section 4

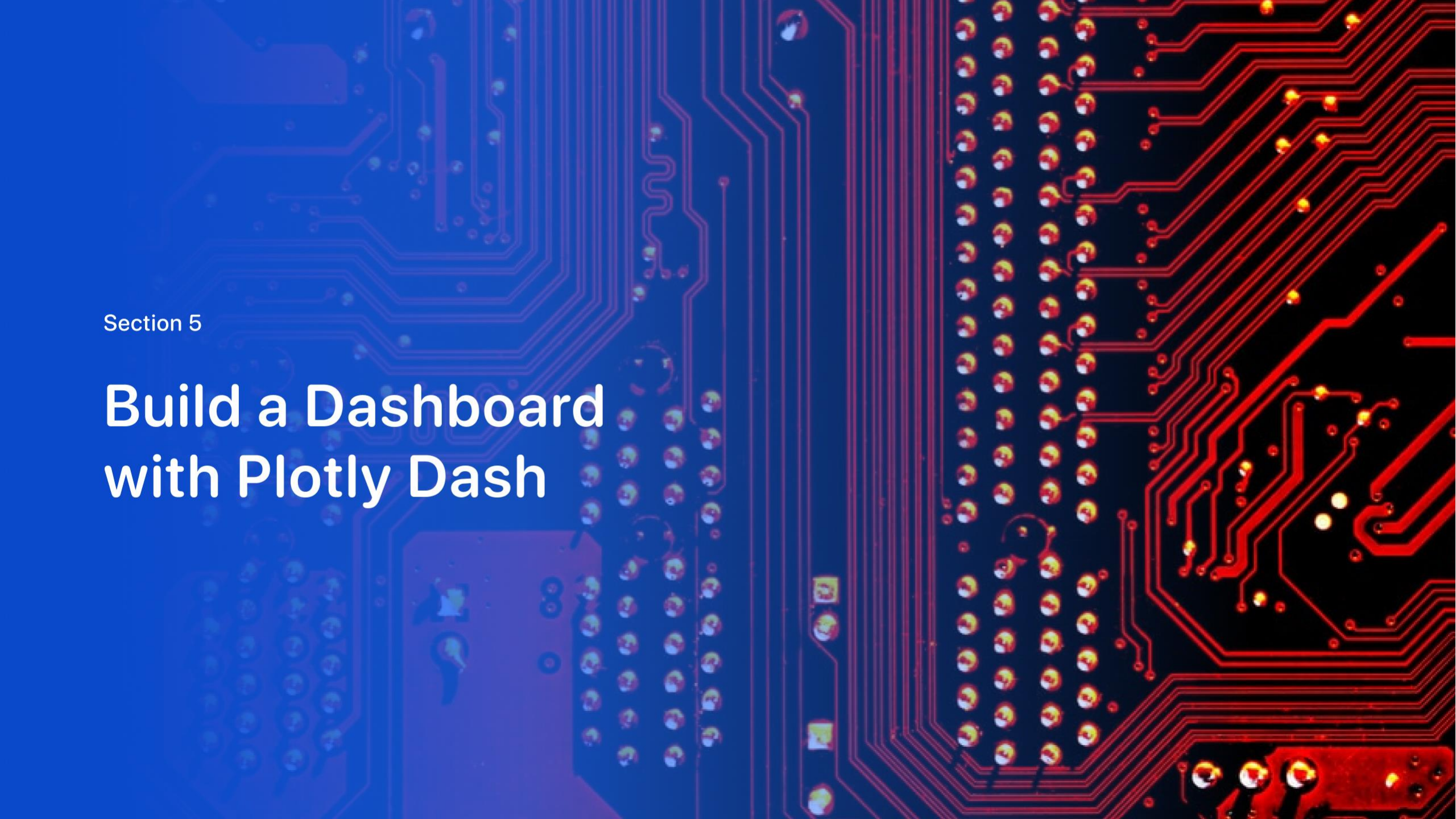
Launch Sites Proximities Analysis



Interactive Visual Analytics (Folium)

- **Water Proximity:** All launch sites are located right on the coast so flight and abort paths stay over open water.
- **Transport Access:** Facilities sit near major roads and rail lines for booster transport while remaining far from cities.
- **Recovery Placement:** Offshore drone ship locations match direct downrange flight paths for each orbit type.
- The link to the notebook is [https://github.com/onurtorun/IBM-Applied-Data-Science-Capstone/blob/main/3-1 Interactive Visual Analytics with Folium lab/jupyter launch site location.ipynb](https://github.com/onurtorun/IBM-Applied-Data-Science-Capstone/blob/main/3-1%20Interactive%20Visual%20Analytics%20with%20Folium%20lab/lab%20jupyter%20launch%20site%20location.ipynb)





Section 5

Build a Dashboard with Plotly Dash

Interactive Dashboard (Plotly Dash)

- **Live Analytics:** Built a dashboard to filter results across launch sites and payload ranges in real time.
- **Best Complex:** KSC LC-39A showed the highest overall landing success rate among all sites.
- **Sweet Spots:** Boosters carrying payloads between 2,000 to 4,000 kg and above 9,000 kg saw consistently high recovery rates.

The link to the notebook is

[https://github.com/onurtorun/IBM-Applied-Data-Science-Capstone/blob/main/3-1 Interactive Visual Analytics with Folium lab/lab_jupyter_launch_site_location.ipynb](https://github.com/onurtorun/IBM-Applied-Data-Science-Capstone/blob/main/3-1%20Interactive%20Visual%20Analytics%20with%20Folium%20lab/lab_jupyter_launch_site_location.ipynb)

Total Success Launches By Site



Section 6

Predictive Analysis (Classification)

Machine Learning Setup

- **Data Preparation:** Applied one-hot encoding to categorical features, creating an 83-feature dataset.
- **Scaling:** Standardized all numeric inputs with StandardScaler to balance feature weights.
- **Train/Test Split:** Divided data into 80% training (72 flights) and 20% testing (18 flights).
- **Models Tested:** Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN).

The link to the notebook is

[https://github.com/onurtorun/IBM-Applied-Data-Science-Capstone/blob/main/4-1 Complete the Machine Learning Prediction lab/SpaceX Machine%20Learning%20Prediction Part 5.ipynb](https://github.com/onurtorun/IBM-Applied-Data-Science-Capstone/blob/main/4-1%20Complete%20the%20Machine%20Learning%20Prediction%20lab/SpaceX%20Machine%20Learning%20Prediction%20Part%205.ipynb)

Decision Tree Evaluation

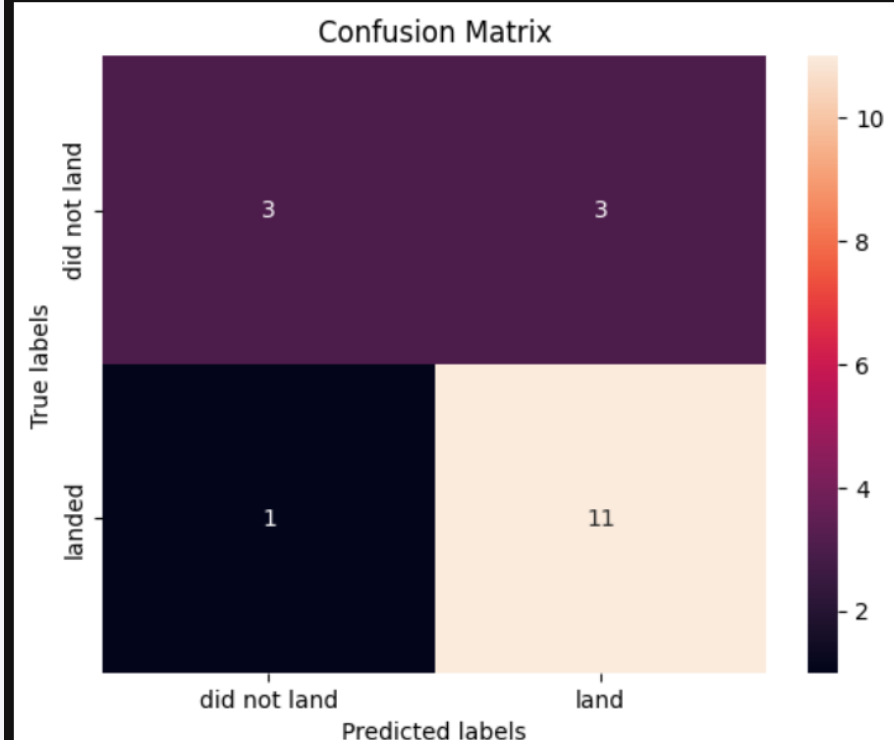
- **Selected Settings:** criterion='entropy', max_depth=14, max_features='sqrt', min_samples_leaf=2, splitter='random'.
- **Training Accuracy:** 85.89% cross-validation score across 10 folds.
- **Test Set Check:** The confusion matrix shows solid precision with minimal false negatives on holdout data.

Calculate the accuracy of tree_cv on the test data using the method `score` :

```
# Evaluate Decision Tree test accuracy
tree_test_score = tree_cv.score(X_test, Y_test)
print("Test set accuracy: ", tree_test_score)

# Predict test labels and plot confusion matrix
yhat = tree_cv.predict(X_test)
plot_confusion_matrix(Y_test, yhat)
```

Test set accuracy: 0.7777777777777778



Model Comparison & Results

- **Test Set Accuracy:** All four tuned models achieved an identical test accuracy of 83.33% (correctly predicting 15 of 18 test flights).
- **Model Behavior:** Logistic Regression and SVM delivered stable decision boundaries, while Decision Tree captured categorical interactions well.
- **Practical Choice:** Given small dataset constraints, simpler models like Logistic Regression offer the lowest risk of overfitting.

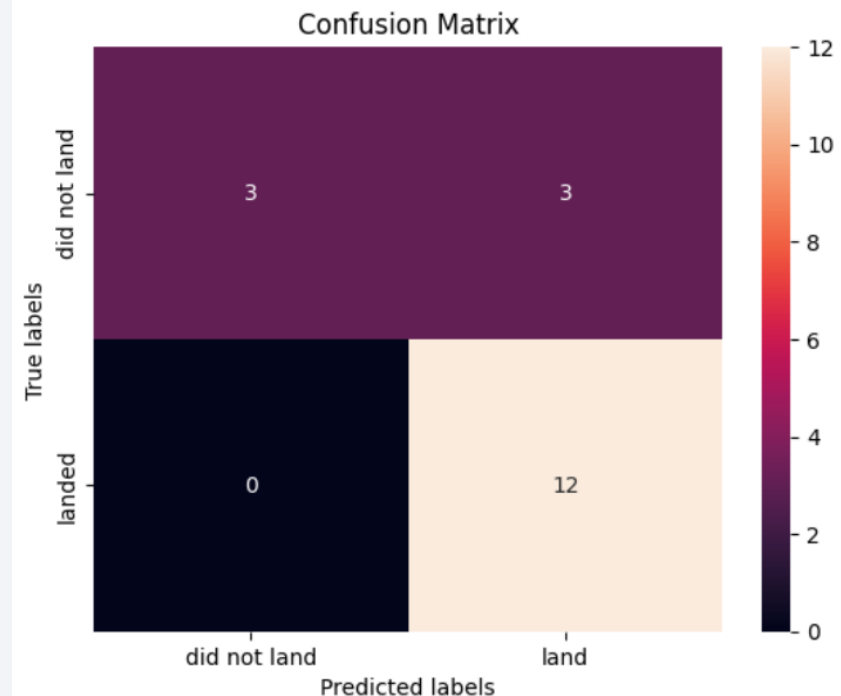
TASK 12

Find the method performs best:

```
# Evaluate KNN test accuracy
knn_test_score = knn_cv.score(X_test, Y_test)
print("Test set accuracy: ", knn_test_score)

# Predict test labels and plot confusion matrix
yhat = knn_cv.predict(X_test)
plot_confusion_matrix(Y_test, yhat)
```

Test set accuracy: 0.8333333333333334



Conclusions & Business Recommendations

- **Commercial Bidding:** Competitors should assume SpaceX will reuse boosters on low-energy LEO and ISS missions and price bids accordingly.
- **Vulnerable Missions:** High-mass GTO launches offer the best opening for competitors, as tight fuel margins make booster recovery far riskier for SpaceX.
- **Industry Direction:** Matching SpaceX's pricing long-term requires developing reusable first stages and sea-recovery platforms.

Conclusions & Business Recommendations

- **Complete Codebase:** All notebooks, datasets, SQL files, and dashboard scripts are hosted on GitHub:

<https://github.com/onurtorun/IBM-Applied-Data-Science-Capstone>

Thank you!

